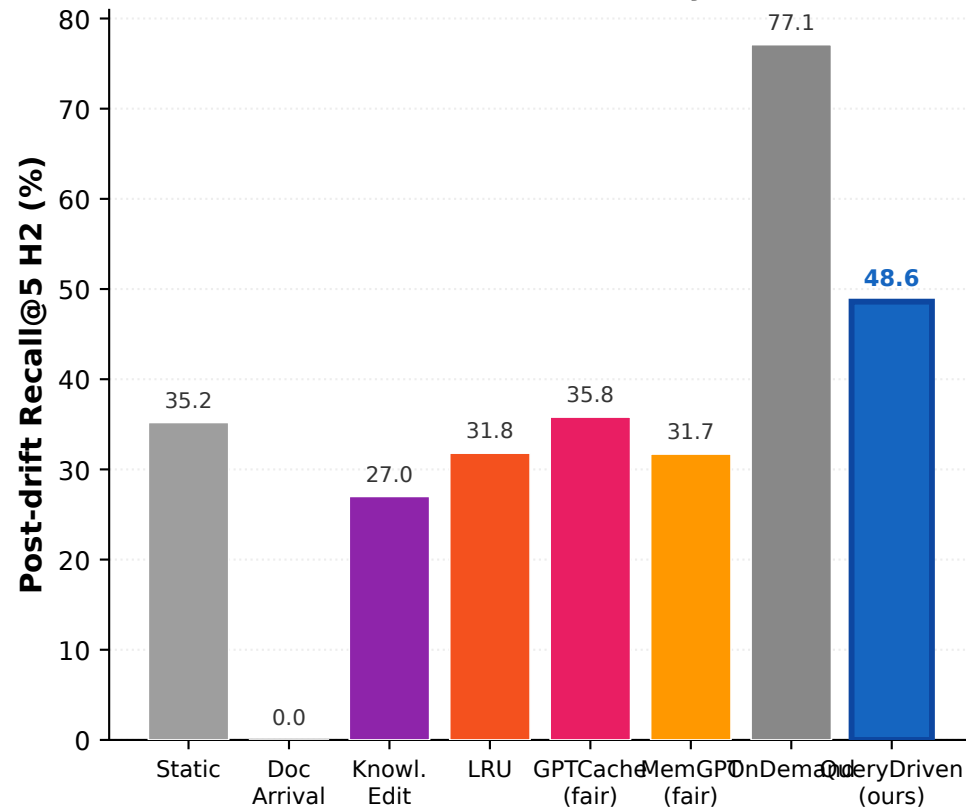
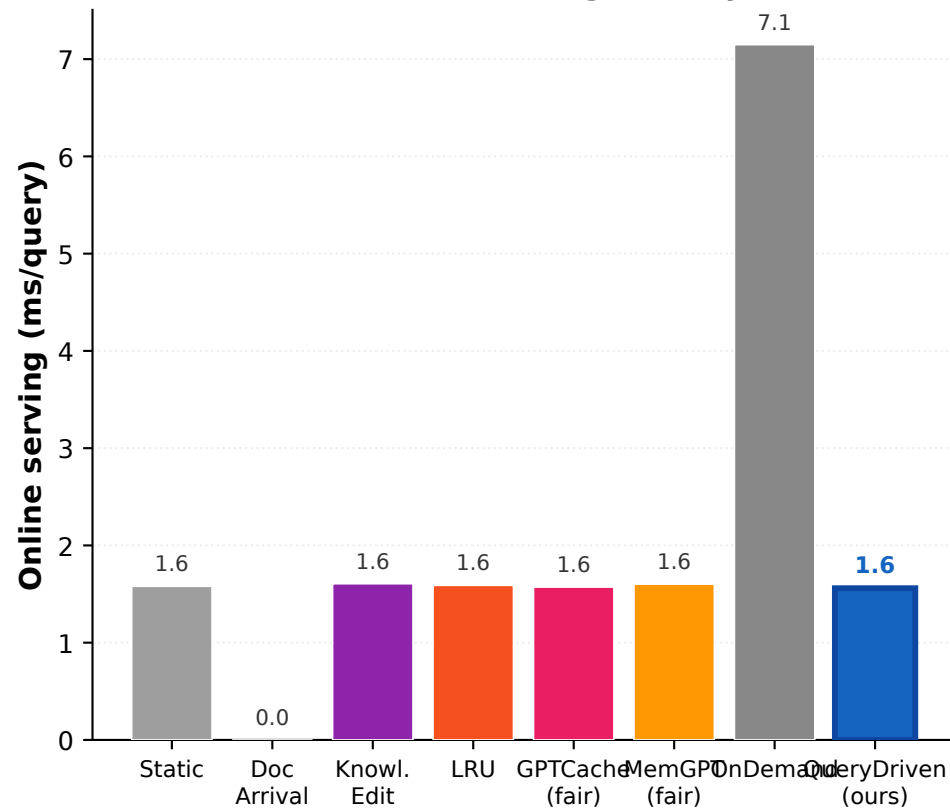


Hot-tier cache quality vs. production-equivalent cost (HotpotQA-comp, pool=200k / KB=21k, gradual drift)
(Encoder: all-MiniLM-L6-v2 · Dense retrieval on HotpotQA-comp)

(a) Retrieval Quality



(b) Online Serving Latency



(c) Offline Update Overhead

